Journal of South American Earth Sciences 104 (2020) 102815

Contents lists available at ScienceDirect

Journal of South American Earth Sciences

journal homepage: www.elsevier.com/locate/jsames

An integrated paleomagnetic and multispecimen paleointensity study from the late Jurassic Zapicán dike swarm (Uruguay)

M. Cervantes-Solano [a,*], A. Goguitchaichvili [b], L. Sánchez Bettucci [c], J. Morales-Contreras [b], Claudia Gogorza [d], Pablo Núñez [c,e]

[a] Laboratorio Interinstitucional de Magnetismo Natural, Escuela Nacional de Estudios Superiores Unidad Morelia, UNAM, Morelia, Mexico
[b] Laboratorio Interinstitucional de Magnetismo Natural, IGUM-UNAM, Morelia, Mexico
[c] Facultad de Ciencias, Universidad de la República, Uruguay
[d] Centro de Investigaciones en Física e Ingeniería del Centro de la Provincia de Buenos Aires, (CIFICEN- CONICET - UNCPBA - CICPBA), Tandil, Argentina
[e] IGEBA, Universidad de Buenos Aires, Buenos Aires, Argentina

ARTICLE INFO

Keywords:
Paleomagnetism
Rock-magnetism
Paleointensity
Uruguay
Dikes
Zapicán

ABSTRACT

A rock magnetic, paleomagnetic and absolute intensity study was conducted on 21 basic dike swarms situated in the Zapicán area (Lavalleja, Uruguay). Paleomagnetic directions revealed mostly normal polarity magnetizations with few intermediate and reverse polarity dikes. Combining data from a previous study, a mean paleodirection obtained is Dec = 3.5°, Inc = −50.6°, α_{95} = 6.3°, N = 19, corresponding to a paleomagnetic pole PLat = 86.4° S, PLong = 178.9° E, A_{95} = 7.2°. This pole position coincides, within uncertainties, with other poles proposed for the same region and age. The paleomagnetic intensities obtained by the multispecimen method including fraction and domain state corrections yielded values from 13.4 to 66.5 μT (and from 2.5 to 12.4 10^{22} Am^2 in terms of virtual axial dipole moments). This indicates that the geomagnetic field probably experienced relatively high fluctuations at the end of the Jurassic Hyperactivity Period (more than 10 reversals per million years) questioning the relationship between the strength of the geomagnetic field and reversal rate.

1. Introduction

The Earth's magnetic field (EMF) is characterized by a continuous change of its direction and intensity. Analysis of the geomagnetic field variation can provide valuable information about the mechanisms that rule the variation of the magnetic field in Earth's interior, providing boundary conditions for the development of geodynamo models. Remanent magnetization, can also provide information about the field intensity in which it was created. Nevertheless, the absolute paleointensity of the EMF cannot be determined from the depositional or post-depositional remanence of sedimentary rocks, but it can only be obtained from the thermoremanent magnetization (TRM) acquired by igneous rocks. The amount of existing paleointensity data is much lower than the amount of directional paleomagnetic data, and their geographical and temporal distribution is still very biased. On the other hand, numerical simulations of the geodynamo suggest that long periods with a stable polarity, such as superchrones, are controlled by the thermal structure of the lower mantle. Various authors have suggested a certain correlation between the intensity of the EMF and the length of the polarity intervals (Smirnov et al., 2017; Doubrovine et al., 2019). However, the absolute intensity determinations in the major part of the Mesozoic including the Cretaceous Normal Superchron have provided contradictory results, with unusually high, low and present day virtual dipole moment values (Kirschvink, 1980; Qin et al., 2011; Qin et al., 2011; Fisher, 1953; Ruiz et al., 2006). Tauxe and Yamazaki, 2007 and Kulakov et al. (2019) revealed that the geographical distribution of palointensity detrminations is characterized by a great asymmetry and existing data does not show a clear long-term trend. Thus, for a reliable description of the temporal variations of the EMF, a less biased knowledge of its characteristics is indispensable.

The tectonic terranes of Piedra Alta and Nico Pérez in Uruguay are separated by the Paleoproterozoic Sarandí del Yí shear zone (Sánchez Bettucci et al., 2010). These areas are intruded by different basic dikes swarms. One of them, the Zapicán dike swarm (Bossi and Schipilov, 2003) emerges near the towns of Nico Peréz and Zapicán, in the department of Lavalleja in Uruguay. Lossada et al. (2014) report the preliminary paleomagnetic results indicating the presence of normal and reverse geomagnetic polarities for this dike swarm. These authors also

* Corresponding author. Escuela Nacional de Estudios Superiores Unidad Morelia, UNAM, Morelia, Mexico.
E-mail address: miguel_cervantes@enesmorelia.unam.mx (M. Cervantes-Solano).

https://doi.org/10.1016/j.jsames.2020.102815
Received 16 April 2020; Received in revised form 28 July 2020; Accepted 10 August 2020
Available online 22 August 2020
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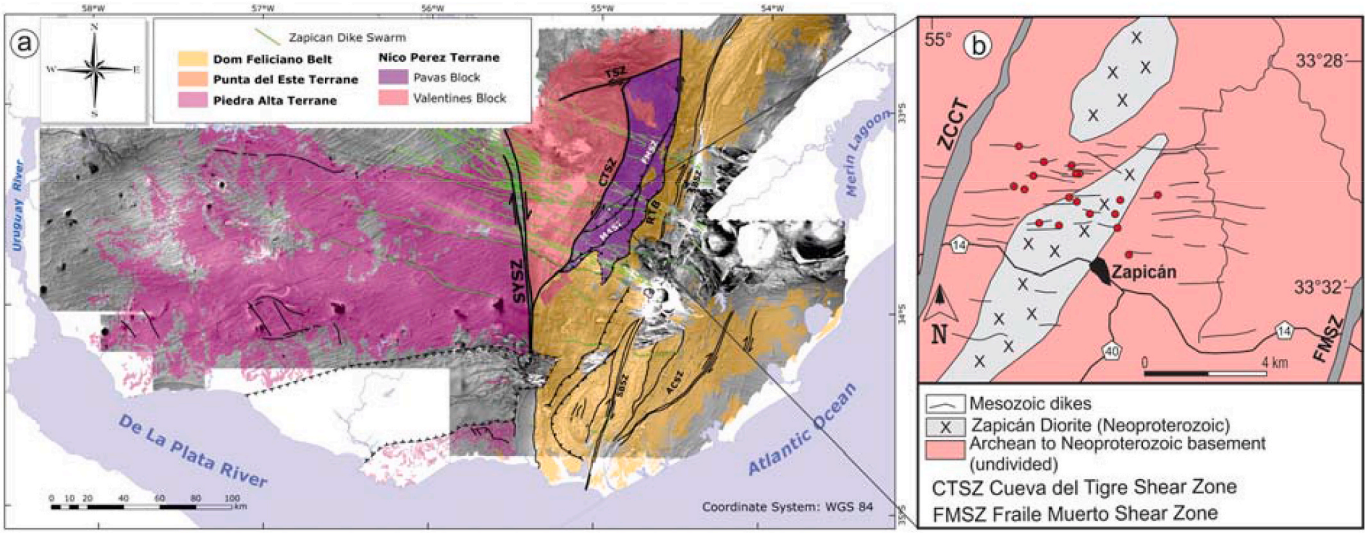


Fig. 1. a) Simplified geological map of the southern Uruguay (modified from Loureiro et al., 2019), b) geological sketch of Zapicán region. Location of the sampled sites in red circles (modified from the Lossada et al., 2014). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

describe these dikes as sub-vertical basaltic structures and with a predominantly E-W to NE-SW orientation providing an isotope age of 157.6 ± 3 Ma. It seems that the main intrusion of the swarm of basic dikes from Nico Pérez-Zapicán occurred in the Mesozoic, most probably associated with the first tectonic efforts that caused the opening of the South Atlantic Ocean.

In order to obtain new paleomagnetic data in the region and obtain a robust paleomagnetic pole position during the late Jurassic, a paleomagnetic-sampling campaign was carried out in the summer of 2015. In addition, the special effort is paid to retrieve absolute geomagnetic intensities to contribute to the Mesozoic Dipole Low controversy and the relationship (if any) between the EMF strength and geomagnetic reversal rate.

2. Geological context and sampling details

The Precambrian units of Uruguay are represented by the Piedra Alta and Nico Pérez terranes, separated by the Sarandí del Yí shear zone (Sánchez Bettucci et al., 2010). Both areas are intruded by different Mesozoic dike bodies, including the Zapicán basic dike swarm (Bossi and Campal, 1991; Teixeira et al., 1999; Bossi et al., 1993; Halls et al., 2001; Bossi and Schipilov, 2003). It mainly outcrops in the surroundings of the towns of Nico Pérez and Zapicán, department of Lavalleja, Uruguay (roughly around 33° 31′S, 55° 56′O). They are essentially basaltic, sub-vertical, and predominantly E-W to NW-SE dikes. The study comprises an area of approximately 30 km long by 10 km wide. The Nico Pérez terrane is divided into Valentines and Pavas Blocks. The Valentines Block is constituted by metatrondhjemites, metagranites and metatonalites, ferriferous conglomerates, BIFs (Bounded Iron Formation), cherts, amphibolites, pyroxenites, garnet-pyroxene quartzites and forsterite-diopside marbles (Bossi and Umpierre, 1969; Masquelin, 2006; among others). The Pavas block is divided into La China and Las Tetas Complexes. This Complex represent the oldest (Archean) rocks from Uruguay (Hartmann et al., 2001; Gaucher et al., 2014). The dikes intrude the Paleoproterozoic and Archean basement of the Valentines Block. In the eastern part, the Zapicán Diorite, intrusive in the Pavas block (La China Complex sensu Hartmann et al., 2001), crops out as a NNE–SSW body (mapped by Lossada et al., 2014). It comprises diorites to granodiorites, often containing hornblende and gabbroic enclaves. Oriolo et al. (2016) obtained an age of 610 Ma for the Zapicán Diorite and is cut by several dikes. Rivalenti et al. (1995) (Moskowitz et al., 1998; Núñez Demarco et al., 2020; Oyhantçabal et al., 2011) presented the first data for this dikes swarm, suggesting Brasiliano age. The age obtained is 665 ± 203 Ma (Rb/Sr), but their Rb–Sr data do not define an isochron, and the error is very large (±203 Ma) and the fit is poor (MSWD = 21.3). The studied dikes was dated by $^{39}Ar/^{40}Ar$ by Lossada et al. (2014) on a geochronological study reported obtaining an absolute age of 157.6 ± 3 Ma 39Ar/40Ar and is considered as the best estimate of the age of intrusion of the dikes available.

Lossada et al. (2014), based on a study of anisotropy of magnetic susceptibility, established that the direction of the magmatic flow within the dikes was predominantly sub-vertical. These dike swarms are associated with continental flood basalts, although their extension and distribution are often difficult to assess due to lack of exposure (Ernst and Buchan, 1997). Based on the aeromagnetic maps, a giant radiated mafic dike swarm is well defined, and was renamed as Nico Pérez-Zapicán Dyke Swarm Group by Núñez et al. (2020).

In total, 21 dikes samples were collected obtaining at least eight standard oriented cores in most of the cases by both magnetic and solar compasses (Fig. 1). Special effort was paid to sample relatively fresh and unaltered outcrops.

3. Experimental procedures

In order to recognize the nature of the minerals responsible for the magnetization, susceptibility vs. temperature experiments (k-T curves) were carried out using an MFK1 magnetic susceptibility meter (AGICO) equipped with by high temperature furnace apparatus (CS-3). Powder sample of each site of approximately 1 g was heated to ca. 700 °C, the heating rate was 15 °C/min and the samples were subsequently cooled down at the same rate. The whole experiment was performed in the presence of argon gas to reduce the possibility of oxidation. Curie points were determined following the method described in Prévot et al. (1983).

Natural remanent magnetization (NRM) of all samples was measured using an AGICO JR6A spinner magnetometer. Subsequently, samples were demagnetized both by heating and alternating fields to determine the characteristic remanent magnetization (ChRM) directions. The demagnetization by alternating fields was performed using 5 to 12 steps applying fields with a peak value of up to 90 mT using an AGICO LDA-3 AF demagnetizer. Thermal treatment was performed with an ASC TD48-SC thermal demagnetizer, reaching temperatures up to 540 °C in 10 heating steps. The ChRM directions of each sample were calculated using Principal Component Analysis (PCA) (Kirschvink, 1980) with at least 5 demagnetization steps. The mean paleomagnetic directions and

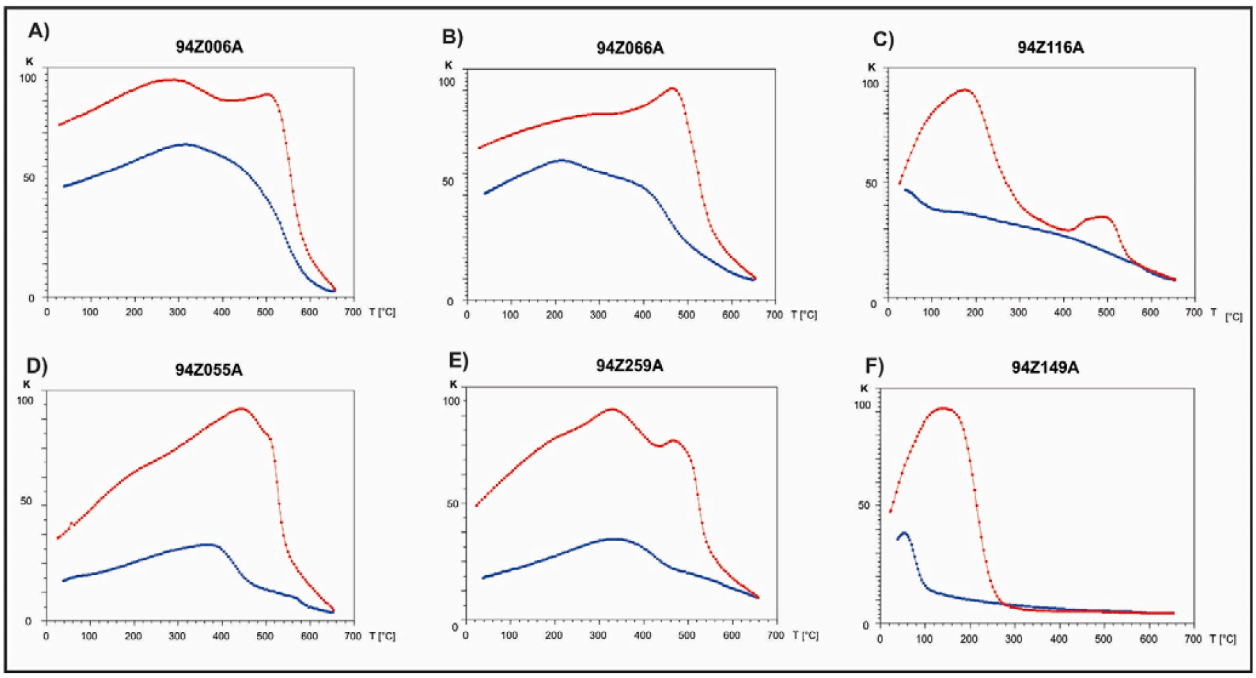


Fig. 2. Representative examples of magnetic susceptibility as a function of temperature (k-T curves). The red lines indicate the behavior of susceptibility during heating and the blue lines during the cooling cycle (arbitrary units). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

their corresponding statistical precision parameters were obtained using Fisher's statistics (Fisher, 1953).

For multispecimen paleointensity experiments, specimens were selected from those that showed a single, well-defined component of magnetization during magnetic treatments and reasonable reversible thermomagnetic curves with a single ferrimagnetic phase (Curie temperatures between 500 and 570 °C). Nine standard paleomagnetic specimens were cut into eight sub-specimens and pressed into salt pellets to obtain standard-dimension cylindrical paleomagnetic specimens (72 in total). Paleointensity experiments were performed with the

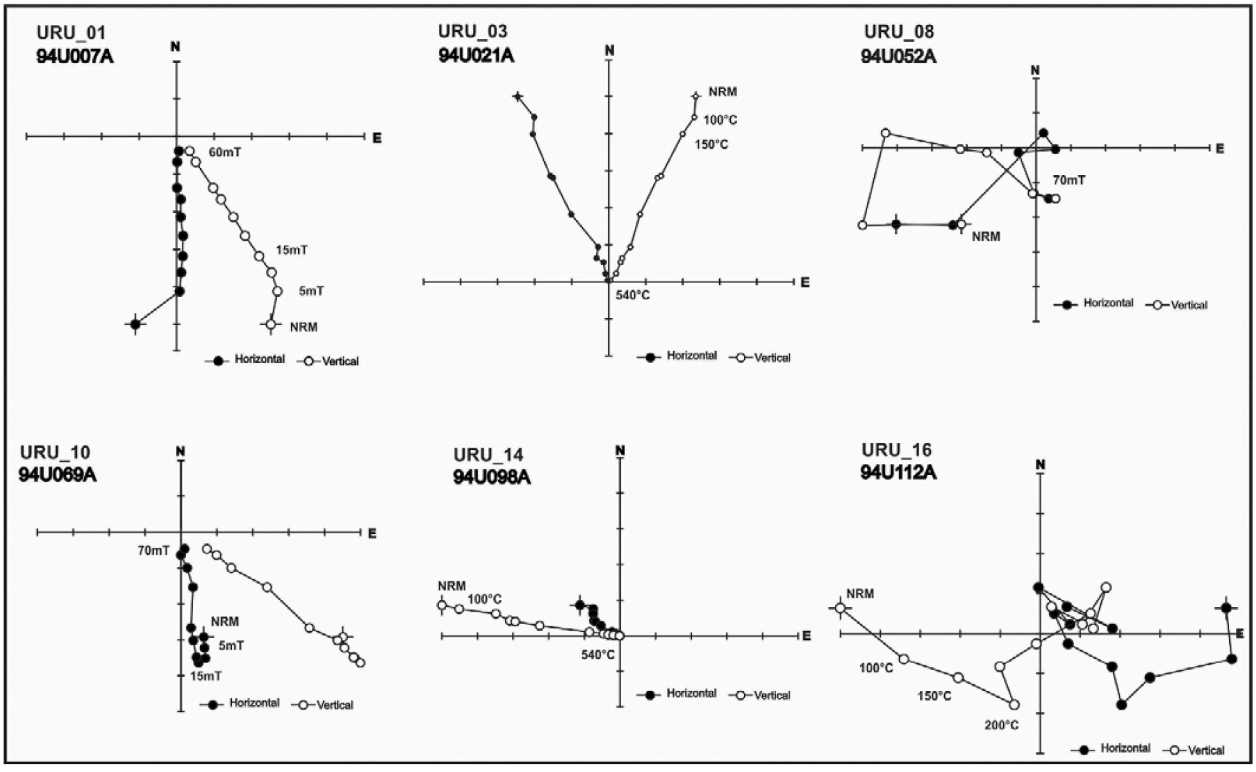


Fig. 3. Representatives orthogonal vector plots of stepwise alternating field or thermal demagnetization. The numbers refer to the peak alternating field or temperature step.

Table 1
Summary of site-mean paleodirections. n is the number of specimens used for calculation; N is the number of treated samples; Inc. inclination; Dec. declination; α_{95} and k radius of 95% confidence angle and precision parameter of Fisher (1953)tatistics; Plat and Plong are the latitude and longitude of the virtual geomagnetic poles respectively (in southern hemisphere coordinates). URUZ sites correspond to this study and ZAP sites correspond to Lossada et al. (2014); ND - not determined. Data in bold are the sites with reverse polarity.

Site	n/N	Dec. [°]	Inc. [°]	α_{95} [°]	K	Plat. [°]	Plong. [°]
URU_01	7/8	186.3	36.7	7.1	73	−75.8	329.8
URU_02	6/8	27.3	−47.6	4.4	303	−66.2	210.9
URU_03	5/7	18.4	−40.9	11.5	45	−71.0	187.9
URU_04	0/7	*ND*	*ND*	*ND*	*ND*	*ND*	*ND*
URU_05	5/7	12.5	−45.5	3.8	590	−77.4	187.2
URU_06	0/7	*ND*	*ND*	*ND*	*ND*	*ND*	*ND*
URU_07	4/7	14.2	−60	3.2	830	−76.5	252.4
URU_08*	4/6	223.9	−0.9	8	132	−36.6	4.8
URU_09	0/7	*ND*	*ND*	*ND*	*ND*	*ND*	*ND*
URU_10	**8/8**	**170.8**	**56.3**	**1.3**	**2250**	**−81.8**	**8.4**
URU_11	0/7	*ND*	*ND*	*ND*	*ND*	*ND*	*ND*
URU_12	5/7	349.1	−53.7	7.1	116	−80.9	27.4
URU_13*	5/7	128.5	−0.9	11.2	69	−31.0	239.2
URU_14	5/7	359.7	−71.2	16.2	23	−67.8	305.5
URU_15	**7/8**	**154**	**54.6**	**14.4**	**23**	**−68.5**	**23.4**
URU_16	0/7	*ND*	*ND*	*ND*	*ND*	*ND*	*ND*
URU_17	6/8	339.8	−29.8	9.5	66	−64.7	73.9
URU_18	0/7	*ND*	*ND*	*ND*	*ND*	*ND*	*ND*
URU_19	0/7	*ND*	*ND*	*ND*	*ND*	*ND*	*ND*
URU_20*	5/7	123.9	−6.3	6.3	147	−25.7	238.1
URU_21	5/8	23.5	−44.3	19.8	22	−68.3	201.1
ZAP_03	**5/9**	**172.4**	**60.8**	**11.9**	**19**	**−79.7**	**338.8**
ZAP_09	7/10	12.6	−50.5	13.9	22	−79.1	206.6
ZAP_13	**6/16**	**178.3**	**65.0**	**9.6**	**50**	**−76.4**	**310.2**
ZAP_15	11/19	12.6	−52.0	7.2	42	−79.4	213.2
ZAP_16	6/15	352.6	−32.5	7.4	81	−72.8	100.3
ZAP_18	5/14	348.5	−38.6	5.3	210	−74.5	81.4
ZAP_20	9/19	26.3	−47.3	6.2	69	−67.0	209.7
ZAP_21	8/14	4.1	−54.0	5.7	94	−86.4	233.3

multispecimen method proposed by Dekkers and Böhnel (2006), including the extended protocols for fraction correction (FC) and domain state correction (DSC) proposed by Fabian and Leonhardt (2010). These experiments were carried out using an ASC Scientific TD48-SC furnace using laboratory fields from 20 to 70 μT, with increments of 10 μT, at a temperature of 480 °C. We chose 480 °C to impart different laboratory fields because more than 65% of initial remanence is removed and thus the segment of NRM used for paleointensity determination is representative. Moreover, at higher temperatures, there is a higher possibility of magneto-chemical modifications as also suggested by thermomagnetic curves.

Calculations were performed through the VBA software implemented by Monster et al. (2015). An acceptable angular deviation (AAD) of 10° and $\alpha = 0.5$ was chosen for the calculations by the VBA software, while the overprint check, the relative alteration error ($\varepsilon_{alt} \leq 20\%$), and the intersection with the y-axis ($\Delta b = -1$ at $H_{lab} = 0$) were all considered as reliability criteria.

4. Results and discussion

The continuous thermomagnetic experiments showed that most of the samples (60%) show two ferrimagnetic phases during heating and only one during cooling (Fig. 2A, C and D). For sample 94Z006A, the first phase appears at temperatures close to 360 °C while the second one reaches approximately 560 °C. This may indicate either transformation of magnetically unstable titanomahgemite or co-existence of both titanomagnetite with low and high titanium content. Another important group of samples (25%) presented a single ferromagnetic phase with a Curie temperature around 560–572 °C, compatible with titanomagnetite with a very low titanium content (Fig. 2B). In the rest of the samples, the presence of a very low Curie temperature phase at around 280 °C (Fig. 2E) is detected which apparently co-exist with a magnetite phase. Alternatively, Curie temperature variation in non-stoichiometry titanomagnetites may be related to the cation substitution/deficiency, vacancy concentration, and its thermal history (Jackson and Bowles, 2018; Bowles et al., 2019).

After measuring the NRM in all available specimens, and after thermal and alternating field demagnetization, it is observed that most of the samples yielded evidence for a single, reasonably stable component of magnetization (Fig. 3, samples 94U007A, 94U021A). In other cases, two components are recognized (Fig. 3, sample 94U069A), the first of them is considered as low coercivity phase and it was possible to remove it at the first steps of demagnetization (5–15 mT or 100 °C), the second component is considered as of high coercivity and presents a linear behavior towards the origin. In a significant number of samples (~20%), incoherent behavior was observed during the demagnetization processes, so it was not possible to determine their main magnetization components and consequently, they were discarded for the site-mean calculation (Fig. 3, samples 94U052A and 94U112A).

After obtaining the directions of the ChRM of the individual specimens, the mean paleomagnetic directions were successfully obtained for 14 of the studied sites, revealing 8 normal, 3 reverse and another 3 of intermediate polarity magnetizations (Table 1, Fig. 4A). The cut-off angle of VGP (Virtual Geomagnetic Pole) for calculating the paleomagnetic pole was set at paleolatitude < 45°. This allowed us to separate stable (normal or reverse) from intermediate geomagnetic regimes). Despite their geomagnetic significance, sites exhibiting intermediate polarities were discarded for the calculation of the paleomagnetic pole-position. Because of few sites provided reverse polarity magnetization, the reversal test of McFadden and McElhinny (1990) yielded an undetermined result.

The mean paleodirection of the studied dikes is Dec = 3.7°, Inc = −50.4°, $\alpha_{95} = 9.5°$, N = 11, corresponding to a paleomagnetic pole of PLat = 86.2° S, PLong = 178.6° E, $A_{95} = 10.2°$ (Fig. 4B). If we consider the well-defined normal and reverse polarity sites reported by Lossada et al. (2014), (ZAP sites in Table 1), a new, combined paleomagnetic mean direction is Dec = 3.5°, Inc = −50.6°, $\alpha_{95} = 6.3°$, N = 19, and the corresponding the paleomagnetic pole of PLat = 86.4° S, PLong = 178.9°

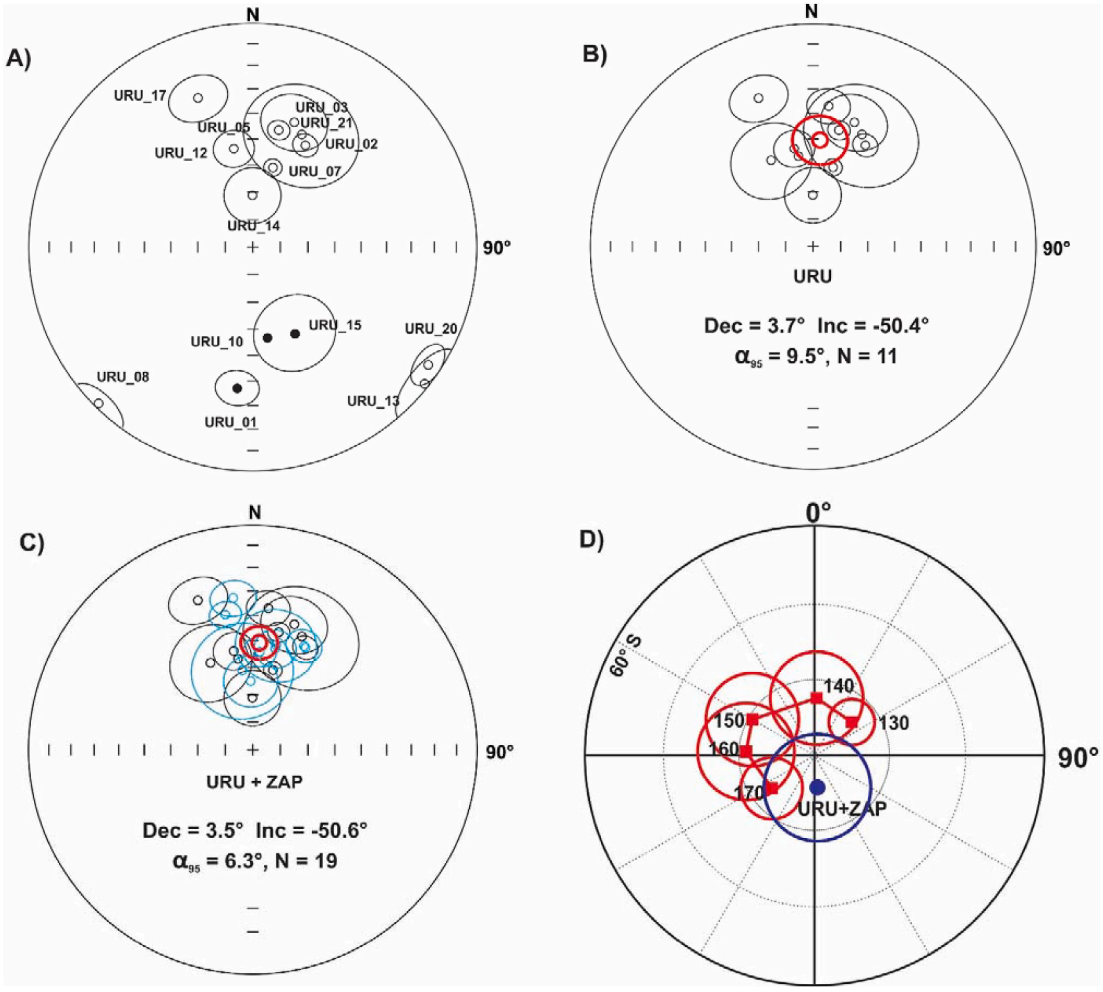


Fig. 4. Equal area projections of the site-mean paleodirections for the Zapicán dikes. Full black circles indicate positive inclinations; open circles indicate negatives inclinations. A) Mean directions of all sites obtained under this investigation and B) mean directions after transformation of reversed antipodal directions and discarding the intermediate polarities, the Fisher mean is plotted in red. C) Combined mean direction including data (blue circles) reported by Lossada et al. (2014) and the mean directions found in this study (black circles). D) Pole positions for this study combined with Lossada et al. (2014) selected data (blue circle), the red circles are adjacent ages poles recalculated from Torsvik et al. (2008), the numbers indicate the pole position age in Ma. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 2
Summary of multispecimen paleointensity determinations including fraction (FC) and domain state (DSC) corrected values. Accepted results are shown in bold (see text for the selection criteria applied).

	Sample	PI [μT]	Min [μT]	Max [μT]	Avg ε_{alt} [%]	Δb	r^2
QDBc	URU-ZA1	112.4	69.7	N.A	−9.3	N.A.	0.822
QFCc	URU-ZA1	192.5	N.A.	N.A.	−9.3	0.02	0.946
QDSCc	URU-ZA1	133.4	N.A.	N.A.	−9.3	−0.01	0.981
QDBc	**URU-ZA3**	16.3	15.9	23.0	−27.0	N.A.	0.829
QFCc	**URU-ZA3**	17.3	16.8	22.8	−27.0	0.15	0.881
QDSCc	**URU-ZA3**	13.4	12.8	19.5	−27.0	0.05	0.915
QDBc	**URU-ZA14**	58.6	46.8	67.3	−30.5	N.A.	0.767
QFCc	**URU-ZA14**	76.2	62.2	N.A.	−30.5	−0.26	0.820
QDSCc	**URU-ZA14**	63.4	53.0	N.A.	−30.5	−0.32	0.838
QDBc	URU-ZA17	206.7	N.A.	N.A.	−42.3	N.A.	0.920
QFCc	URU-ZA17	236.5	N.A.	N.A.	−42.3	−0.01	0.923
QDSCc	URU-ZA17	170.9	N.A.	N.A.	−42.3	−0.02	0.940
QDBc	URU-ZA18	89.4	N.A.	N.A.	−36.6	N.A.	0.993
QFCc	URU-ZA18	96.7	N.A.	N.A.	−36.6	−0.20	0.999
QDSCc	URU-ZA18	74.2	N.A.	N.A.	−36.6	−0.27	1.000
QDBc	**URU-ZA20**	84.8	78.1	88.8	−20.3	N.A.	0.979
QFCc	**URU-ZA20**	103.3	N.A.	N.A.	−20.3	0.21	0.987
QDSCc	**URU-ZA20**	66.5	62.8	69.5	−20.3	0.29	0.987

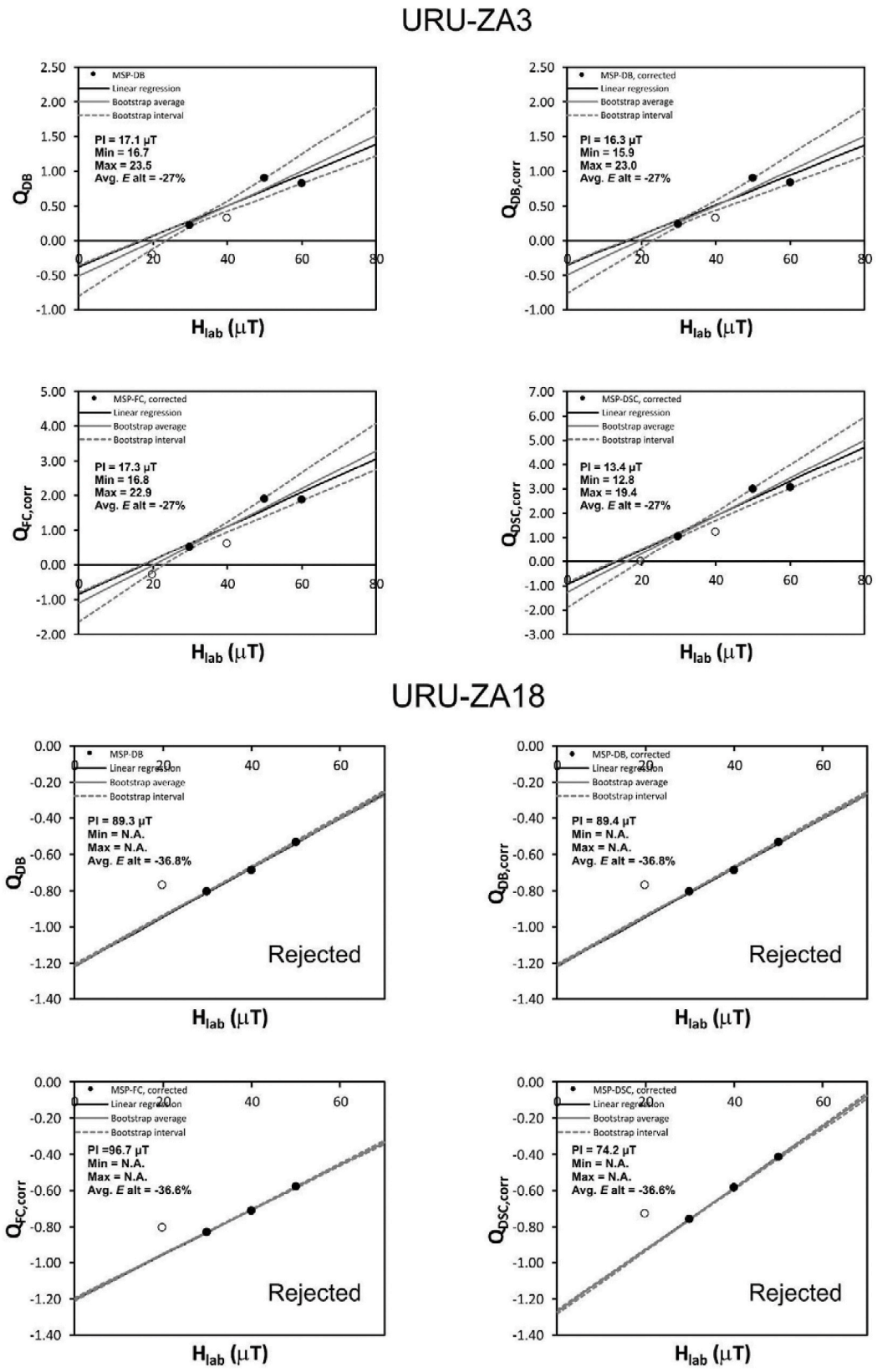


Fig. 5. a. Results of successful paleointensity experiments obtained by the multispecimen methodology. The continuous black lines indicate the linear regression used to obtain the value of the PI, the dashed lines indicate the zone of confidence obtained through the bootstrap type statistical fit. b. Results of unsuccessful paleointensity experiments obtained by the multispecimen methodology. The continuous black lines indicate the linear regression used to obtain the value of the PI, the dashed lines indicate the zone of confidence obtained through the bootstrap type statistical fit.

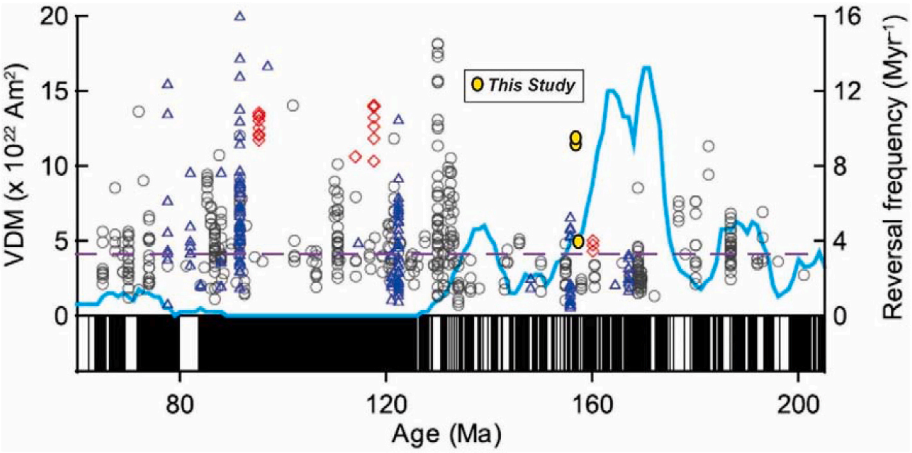


Fig. 6. The latest selected absolute paleointensity compilation reported in Kulakov et al. (2019) between 200 and 65 Ma. Bulk rock data are referred by grey circles, submarine basaltic glass (blue triangles), and single silicate crystals (red diamonds). The light blue curve shows the average reversal frequency calculated with a 5-Ma sliding window (retrieved from Kulakov et al., 2019). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

E, $A_{95} = 7.2°$ (Fig. 4C and D).

This pole position is similar to that obtained in the original study by Lossada et al. (2014) as can be seen in Fig. 4D. The combined paleomagnetic pole agrees well with reference pole position retrieved from Torsvik et al. (2008) for the adjacent age intervals for stable South America. The presence of reverse and intermediate polarities corroborates the hypothesis proposed by Lossada et al. (2014), that the intrusion of the dikes occurred over a relatively extended period and well before the beginning of the Cretaceous Normal Superchron. In this context, the secular variation seems to be sufficiently averaged by the dikes.

Reliable multispecimen paleointensity results were obtained in three of the nine flows studied. For the other three flows, Δb values > -1 were obtained, which suggests that significant alteration ($\varepsilon_{alt} > 30\%$), domain-state effects or misalignment of specimens took place. Multispecimen (MS): paleointensity results for each studied flow, along with associated quality parameters are listed in Table 2, while representative paleointensity plots are shown in Fig. 5a. Accepted values range from 13.4 to 66.5 mT, while the corresponding Virtual Axial Dipole Moment varies from 2.5 to 12.4 10^{22} Am^2. Important paleointensity fluctuation (about 5 times) is detected in present study. However, this fact is not exclusive of Zapican dikes and same amplitude variation is found worldwide (see compilation in Tauxe et al., 2013 as an example). Different interpretations may be suggested involving paleointensity techniques, magnetic mineralogy and true geomagnetic phenomena related to the geodinamo regime. However, no firm conclusion may be reached at this stage of knowledge.

Three individual paleointensity determinations are apparently not enough to make any firm conclusions about the long-term variation of the geomagnetic field strength. However, because of the experimental difficulty involved in determining absolute paleointensities, the number of existing determinations is much less than directional data. Its geographical distribution is also characterized by great asymmetry. About 90% of the paleointensity data contained in the PINT database came from the Northern Hemisphere and 80%, from latitudes above 30°N (Perrin and Shcherbakov, 1997). The PINT06 database used in the analysis performed by Tauxe and Yamazaki (2007) point out that there is still very little data from South America and Africa. Also, the temporal distribution of available data is still very uneven: about 50 data per million years in the period between 1 and 20 million years, and 3 data per million years for the period between 20 and 400 Ma. The number of determinations obtained in older units is even lower. Under these circumstances, three new, paleointensity individual determinations should be considered a valuable contribution despite its preliminary nature.

The geomagnetic field strength during the Jurassic and most of the Cretaceous periods are still a matter of debate (Tauxe et al., 2013). The so-called Mesozoic dipole low (MDL) hypothesis was first introduced by Prévot et al. (1990), revealing persistent low field intensity during most of the Mesozoic time, mainly from 180 to 120 Ma (see also Goguitchaichvili et al., 2002a, 2002b). Perrin and Sherbakov (1997), Selkin and Tauxe (2000), Zhu et al. (2001), among many other authors, basically reinforced this finding. Selkin and Tauxe (2000) argued, however, that the entire time interval from the 160 Ma to Plio-Pleistocene is characterized by a reduced field strength compared to the present-day values. Contrary to this general agreement, Tarduno et al. (2001) reported unusually high paleointensities for the interval 114–116 Ma, within the Cretaceous Normal Superchron. Later, Goguitchaichvili et al. (2002a) reported relatively high paleointensities from 132 Ma old Parana magmatic Province, reporting some values even close to the present-day strength, questioning the existence of the MDL. Tauxe et al. (2013) based on submarine basaltic glass samples advocated for low-field strengths for the entire Jurassic extending into the early Cretaceous, in clear agreement to the initial results suggested by Prévot et al. (1990). Thus, a controversy about the existence of the MDL still persists.

Recently, Kulakov et al. (2019) carried out the most comprehensive paleointensity data analysis from 65 to 200 Ma revealing significant fluctuations in the geomagnetic field strength with respect to the reversal-rate (Fig. 6). The authors clearly demonstrated an inverse correlation between the paleointensity and reversal frequency expressed by the consistently strong median VDM values observed for the Cretaceous Normal Superchron (CNS), and the relatively low median virtual dipole moments values for the Jurassic Hyperactivity Period (JHAP). Our data, on the onset of Jurassic-Cretaceous boundary, indicate the presence of both low and high absolute geomagnetic intensity as may be seen in Fig. 6, and thus, may question the recently reported relationship between geomagnetic strength and polarity reversal rate.

5. Concluding remarks

- 171 samples belonging to 21 dikes were paleomagnetically analyzed. Reliable paleomagnetic directions were obtained from 14 sites; eight among them yielded normal, 3 reverse and 3 intermediate magnetic polarities.
- Magnetic mineralogy is dominated by the coexistence of two ferrimagnetic phases during the heating cycle (Curie temperatures around 360 °C and 560 °C), while only Ti-poor titanomagnetite (almost magnetite) is present during the cooling. Curie temperature variation in non-stoichiometry titanomagnetites may be related to the cation substitution/deficiency, vacancy concentration, and its thermal history (Jackson and Bowles, 2018; Bowles et al., 2019).
- The mean paleodirection of the studied dikes is Dec = 3.7°, Inc = −50.5°, $\alpha_{95} = 9.5°$, N = 11, corresponding paleomagnetic pole of PLat = 86.2° S, PLong = 178.6° E, $A_{95} = 10.2°$, Considering the well-defined normal and reverse polarity sites reported by Lossada et al. (2014), a new, combined paleomagnetic mean direction is Dec =

3.5°, Inc = −50.6°, α_{95} = 6.3°, N = 19, corresponding to a paleomagnetic pole of PLat = 86.4° S, PLong = 178.9° E, A_{95} = 7.2°.
- The combined paleomagnetic pole agrees well with the reference pole position retrieved from Torsvik et al. (2008) for the adjacent age intervals.
- 48 specimens coming from six dikes were subjected for multispecimen paleointensity determination including fraction and domain state corrections. Only three dikes yielded technically acceptable results, with intensities ranging from 13.4 to 66.5 μT corresponding to VADM values between 2.5 and 12.4 10^{22} Am^2.
- Absolute paleointensities obtained in this study contrast the evidence of consistent low field during the so-called Jurassic Hyperactive Period and question the relationship between the geomagnetic strength and polarity reversal rate.

Author statement

We explicitly declare that this manuscript is original, has not been published before and is not currently being considered for publication elsewhere. We confirm that the manuscript has been read and approved by all co-authors. We further confirm that the order of authors listed in the manuscript has been approved by all authors. This work is leaded by Miguel Cervantes-Solano and Avto Goguitchaichvili. All authors participated in the field trip, laboratory measurements and data analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The study was supported by UNAM -PAPIIT grant n° IN101920.

References

Bossi, J., Schipilov, A., 2003. Rocas básicas precámbricas del Uruguay. Rev. Soc. Urug. Geol. 10, 2–14.
Bossi, J., Campal, N., 1991. Granitos Negros Filonianos del Uruguay: Resultados de las investigaciones. Agreement CIID-IDRC – UdelaR, Project N° 90650. Final Report, Ottawa, Canada, p. 72.
Bossi, J., Campal, N., Civetta, L., Demarchi, G., Girardi, V.A.V., Mazzucchelli, M., Teixeira, W., 1993. Early Proterozoic dike swarms from western Uruguay: geochemistry, Sr Nd isotopes and petrogenesis. Chem. Geol. 106 (3–4), 263–277.
Bowles, J.A., Lappe, S.-C.L.L., Jackson, M.J., Arenholz, E., van der Laan, G., 2019. Curie temperature enhancement and cation ordering in titanomagnetites: evidence from magnetic properties, XMCD, and Mössbauer spectroscopy. G-cubed 20, 2272–2289. https://doi.org/10.1029/2019GC008217.
Dekkers, M.J., Böhnel, H.N., 2006. Reliable absolute palaeointensities independent of magnetic domain state. Earth Planet Sci. Lett. 248, 508–517.
Doubrovine, P.V., Velkolainen, T., Pesonen, L.J., Piispa, E.J., Siim, O., Smirnov, A.V., Kulakov, E.V., Biggin, A.J., 2019. Latitude dependence of geomagnetic paleosecular variation and its relation to the frequency of magnetic reversals: observations from the Cretaceous and Jurassic. G-cubed 20, 1240–1279.
Ernst, R.E., Buchan, K.L., 1997. Giant radiating dyke swarms: their use in identifying pre-Mesozoic large igneous provinces and mantle plumes. Geophysical Monograph-American Geophysical Union 100, 297–334.
Fabian, K., Leonhardt, R., 2010. Multiple-specimen absolute paleointensity determination: an optimal protocol including pTRM normalization, domain-state correction, and alteration test. Earth Planet Sci. Lett. 297, 84–94.
Fisher, R.A., 1953. Dispersion on a sphere. Proc. Roy. Soc. Lond. Math. Phys. Sci. 217 (1130), 295–305.
Goguitchaichvili, A., Alva-Valdivia, L.M., Urrutia, J., Morales, J., Ferreira Lopes, O., 2002a. On the reliability of Mesozoic dipole low: new absolute paleointensity results from Paraná flood basalts (Brazil). Geophys. Res. Lett. 29 (13), 33–41.
Goguitchaichvili, A., Urrutia-Fucugauchi, J., Alva-Valdivia, L., 2002b. Mesozoic dipole low: myth or reality? Eos. Trans. Am. Geophys. Union 83 (41), 457–461.
Halls, H.C., Campal, N., Davis, D.W., Bossi, J., 2001. Magnetic studies and U-Pb geochronology of the Uruguayan dyke swarm, Rio de la Plata craton, Uruguay: paleomagnetic and economic implications. J. S. Am. Earth Sci. 14 (4), 349–361.
Jackson, M., Bowles, J., 2018. Malleable Curie temperatures of natural, titanomagnetites: occurrences, modes, and mechanisms. J. Geophys. Res.: Solid Earth 123, 921–940. https://doi.org/10.1002/2017JB015193.
Kirschvink, J.L., 1980. The least-squares line and plane and the analysis of palaeomagnetic data. Geophys. J. Int. 62 (3), 699–718.
Kulakov, E.V., Sprain, C.J., Doubrovine, P.V., Smirnov, A.V., Paterson, G.A., Hawkins, L., Biggin, A.J., 2019. Analysis of an updated Paleointensity database (QPI-PINT) for 65–200 Ma: implications for the long-term history of dipole moment through the Mesozoic. J. Geophys. Res.: Solid Earth 124 (10), 9999–10022.
Lossada, A.C., Rapalini, A.E., Sánchez Bettucci, L., 2014. Enjambre de diques básicos de Nico Pérez-Zapicán, Uruguay: evidencias radimétricas y paleomagnéticas sobre su edad. Rev. Asoc. Geol. Argent. 71 (3), 345–355.
Loureiro, J., Silva, H., Sánchez Bettucci, L., 2019. Mapa Geológico: el Arqueano en el Uruguay. Revista Investigaciones, Montevideo 2 (1), 28–35.
McFadden, P.L., McElhinny, M.W., 1990. Classification of the reversal test in palaeomagnetism. Geophys. J. Int. 103 (3), 725–729.
Monster, M.W., de Groot, L.V., Dekkers, M.J., 2015. MSP-Tool: a VBA-based software tool for the analysis of multispecimen paleointensity data. Front. Earth Sci. 3, 86.
Moskowitz, B.M., Jackson, M., Kissel, C., 1998. Low-temperature magnetic behavior of titanomagnetites. Earth Planet Sci. Lett. 157 (3–4), 141–149.
Núñez Demarco, P., Masquelin, H., Prezzi, C., Muzio, R., Loureiro, J., Peel, E., Campal, N., Sánchez Bettucci, L., 2020. Aeromagnetic patterns in Southern Uruguay: Precambrian-Mesozoic dyke swarms and Mesozoic rifting structural and tectonic evolution. Tectonophysics 228373.
Oriolo, S., Oyhantçabal, P., Basei, M.A.S., Wemmer, K., Siegesmund, S., 2016. The Nico Pérez terrane (Uruguay): from archean crustal growth and connections with the Congo craton to late neoproterozoic accretion to the río de la Plata craton. Precambrian Res. 280, 147–160.
Oyhantçabal, P., Siegesmund, S., Wemmer, K., 2011. The Río de la Plata Craton: a review of units, boundaries, ages and isotopic signature. Int. J. Earth Sci. 100 (2–3), 201–220.
!"count(.//sb:host[1]//child::*//sb:date)">Perrin, M., Shcherbakov, V., . Paleointensity of the Earth's magnetic field for the past 400 Ma: evidence for a dipole structure during the Mesozoic low. J. Geomagn. Geoelectr. 49 (4), 601–614.
Prévot, M., Derder, M.E.M., McWilliams, M., Thompson, J., 1990. Intensity of the Earth's magnetic field: evidence for a Mesozoic dipole low. Earth Planet Sci. Lett. 97 (1–2), 129–139.
Prévot, M., Mankinen, E.A., Grommé, S., Lecaille, A., 1983. High paleointensities of the geomagnetic field from thermomagnetic studies on rift valley pillow basalts from the Mid-Atlantic Ridge. J. Geophys. Res.: Solid Earth 88 (B3), 2316–2326.
Qin, H.F., He, H.Y., Liu y, Q.S., Cai, S.H., 2011. Paleointensity just at the onset of the Cretaceous normal superchron. Phys. Earth Planet. In. 187, 199–211.
Ruiz, R.C., Goguitchaichvili, A., Geuna, S.E., Alva-Valdivia, L.M., Solé, J., Morales, J., 2006. Early Cretaceous absolute geomagnetic paleointensities from Córdoba Province (Argentina). Earth Planets Space 58, 1333–1399.
Sánchez Bettucci, L., Peel, E., Oyhantçabal, P., 2010. Precambrian geotectonic units of the Río de La Plata craton. Int. Geol. Rev. 52 (1), 32–50.
Selkin, P.A., Tauxe, L., 2000. Long-term variations in palaeointensity. Philos. Trans. R. Soc. London, Ser. A: Mathematical, Physical and Engineering Sciences 358 (1768), 1065–1088.
Smirnov, A.V., Kulakov, E.V., Foucher, M.S., Bristol, K.E., 2017. Intrinsic paleointensity bias and the long-term history of the geodynamo. Science Advances 3, e1602306. https://doi.org/10.1126/sciadv.1602306.
Tarduno, J.A., Cottrell, R.D., Smirnov, A.V., 2001. High geomagnetic intensity during the mid-Cretaceous from Thellier analyses of single plagioclase crystals. Science 291 (5509), 1779–1783.
Tauxe, L., Gee, J.S., Steiner, M.B., Staudigel, H., 2013. Paleointensity results from the Jurassic: new constraints from submarine basaltic glasses of ODP Site 801C. G-cubed 14 (10), 4718–4733.
Teixeira, W., Renne, P.R., Bossi, J., Campal, N., D'Agrella Filho, M.S., 1999. ^{40}Ar–^{39}Ar and Rb–Sr geochronology of the Uruguayan dyke swarm, Río de la Plata Craton and implications for Proterozoic intraplate activity in western Gondwana. Precambrian Res. 93 (2), 153–180.
Torsvik, T.H., Müller, R.D., Van der Voo, R., Steinberger, B., Gaina, C., 2008. Global plate motion frames: toward a unified model. Rev. Geophys. 46 (3).
Zhu, R., Pan, Y., Shaw, J., Li, D., Li, Q., 2001. Geomagnetic palaeointensity just prior to the Cretaceous normal superchron. Phys. Earth Planet. In. 128 (1–4), 207–222.